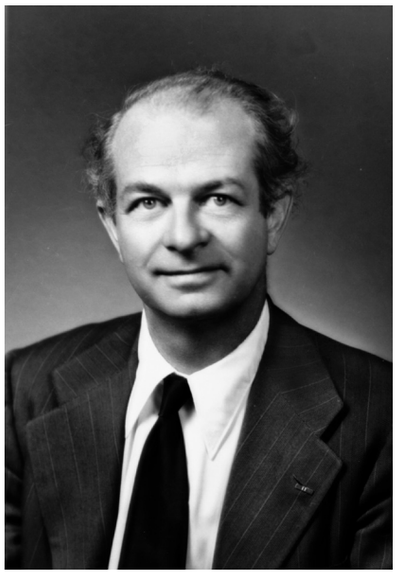
Unit 9 MINERALS

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Linus Pauling

Linus Carl Pauling was an American chemist, biochemist, peace activist, author and educator. He was recognised as one of the 20 greatest scientists of all time. In addition to his contribution to chemistry and he also worked with many biologists. He demonstrated that the hemoglobin molecule changes structure when it gains or loses an oxygen atom. He also uncovered that the sickle cell anemia is caused by an abnormal protein. His success with sickle cell anemia led Pauling to speculate that a number of other diseases, including mental illnesses such as schizophrenia, might result from flawed genetics.

Learning Objectives

After studying this Unit, students will be able to

- Distinguish between the micro and macro nutrients.
- Know the various sources of micro and macro nutrients.
- Appreciate the biological functions of minerals.
- Understand the deficiency causes of various minerals.

Introduction

Minerals are inorganic elements that are present in body fluids and tissues. They are supplied to the body through diet, for example sodium from table salt and sulfur from proteins. They are required for a variety of biochemical and physiological functions such as enzyme action, nerve impulse transmission and muscle contraction. Each mineral is required in a specific amount ranging from micrograms to grams per day.

Minerals constitute about eight percent of total body mass of a human being. Unlike carbohydrates, fats and proteins, minerals donot furnish energy but they influence the rate of biochemical reactions through enzymes. Unlike vitamins, minerals are not destroyed during cooking process. However, some loss may occur due to their solubility in water.

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